The measuredtex package*

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Abstract

The measuredtex package provides semantic markup for measurement models. It wraps standard LaTeX maths environments (`equation`, `align`) so that metadata about each model equation and its constituent terms are written to a sidecar file (`\jobname.models`) during compilation.

Contents

1 Introduction

When writing measurement-model documents it is useful to capture structured metadata alongside the typeset equations. The measuredtex package does this transparently: the author writes equations using familiar environments and the package emits a sidecar file.

*This document corresponds to measuredtex v0.1, dated 2026-06-02.

2 Usage

Load the package in the usual way:

```
\usepackage{measuredtex}
```

2.1 Model environments

`mdeq` (*env.*) A wrapper for `equation`. An optional key–value argument accepts the keys `name`, `role`, `source`, and `label`.

```
\begin{mdeq}[name=Ohm's law, label=eq:ohm]
  V = IR
\end{mdeq}
```

`mdalign` (*env.*) A wrapper for `align`, with the same metadata keys.

2.2 Term definitions

`\mddefn` `\mddefn{`⟨*label*⟩`}{`⟨*term*⟩`}{`⟨*definition*⟩`}` typesets ⟨*term*⟩ in inline maths mode and writes a TERM record to the sidecar file. A warning is issued if ⟨*label*⟩ has not been defined by a prior `\label` command.

2.3 Semantic term macros

`\nspace` `\nspace{`⟨*text*⟩`}` renders a namespace identifier in upright roman type.

`\assoc` `\assoc{`⟨*text*⟩`}` renders an association identifier in upright roman type.

`\tv` `\tv{`⟨*text*⟩`}` renders a term value in italic type.

`\qual` `\qual{`⟨*text*⟩`}` or `\qual{`⟨*text*⟩`}(`⟨*sub*⟩`)` renders a qualifier, optionally followed by a parenthesised sub-qualifier, both in upright roman type.

`\mdterm` `\mdterm[`⟨*ns*⟩`]{`⟨*sym*⟩`}[`⟨*qual*⟩`](`⟨*assoc*⟩`)` composes a decorated term with optional namespace superscript, qualifier subscript, and association argument.

3 Sidecar file format

The file `\jobname.models` contains blocks delimited by header lines:

`===== MODEL =====` starts a model record, followed by key–value lines for `name`, `role`, `source`, `label`, and `equation`.

`--- TERM ---` starts a term record, followed by `eqref`, `term`, and `definition`.

4 Implementation

```
⟨*package⟩
\NeedsTeXFormat{LaTeX2e}
\ProvidesExplPackage {measuredtex} {2026-06-02} {0.1}
  {Semantic markup for measurement models}

\RequirePackage { amsmath }
```

4.1 Messages

```
\msg_new:nnn { measuredtex } { undefined-label }
  { mddefn:~label~'#1'~is~not~defined. }
```

4.2 Metadata variables

```
\tl_new:N \l_md_name_tl
\tl_new:N \l_md_role_tl
\tl_new:N \l_md_source_tl
\tl_new:N \l_md_label_tl
\tl_new:N \l_md_model_tl
```

4.3 Output stream

```
\iow_new:N \g_md_iow

\AtBeginDocument
  {
    \iow_open:Nn \g_md_iow { \jobname .models }
  }

\AtEndDocument
  {
    \iow_close:N \g_md_iow
  }
```

4.4 Metadata keys

```
\keys_define:nn { measuredtex / model }
  {
    name   .tl_set:N = \l_md_name_tl ,
    role   .tl_set:N = \l_md_role_tl ,
    source .tl_set:N = \l_md_source_tl ,
    label  .tl_set:N = \l_md_label_tl ,
  }

\keys_define:nn { measuredtex / mddefn }
  {
    model .tl_set:N = \l_md_model_tl ,
  }
```

4.5 Model-writing logic

`\md_model_write:nn` Clears all metadata keys, applies the supplied key–value pairs, and writes a MODEL block to the sidecar file.

```
\cs_new:Npn \md_model_write:nn #1 #2
  {
    \tl_clear:N \l_md_name_tl
    \tl_clear:N \l_md_role_tl
    \tl_clear:N \l_md_source_tl
    \tl_clear:N \l_md_label_tl

    \keys_set:nn { measuredtex / model } { #1 }

    \iow_now:Nn \g_md_iow { ===== ~ MODEL ~ ===== }

```

```
    \tl_if_empty:NF \l_md_name_tl
      {
        \iow_now:Nx \g_md_iow { name: ~ \tl_use:N \l_md_name_tl }
      }

    \tl_if_empty:NF \l_md_role_tl
      {
        \iow_now:Nx \g_md_iow { role: ~ \tl_use:N \l_md_role_tl }
      }

    \tl_if_empty:NF \l_md_source_tl
      {
        \iow_now:Nx \g_md_iow { source: ~ \tl_use:N \l_md_source_tl }
      }

    \tl_if_empty:NF \l_md_label_tl
      {
        \iow_now:Nx \g_md_iow { label: ~ \tl_use:N \l_md_label_tl }
      }

    \iow_now:Nx \g_md_iow { equation: ~ \tl_to_str:n {#2} }
    \iow_now:Nn \g_md_iow { }
  }
```

(*End of definition for* `\md_model_write:nn`*. This function is documented on page* **??**.)

4.6 Document environments

`mdeq`

```
\NewDocumentEnvironment { mdeq } { O{} +b }
  {
    \md_model_write:nn {#1} {#2}
    \begin{equation}
      #2
      \tl_if_empty:NF \l_md_label_tl
        {
          \label { \tl_use:N \l_md_label_tl }
        }
    \end{equation}
  }
  { }
```

(*End of definition for* `mdeq`*. This function is documented on page* **??**.)

`mdalign`

```
\NewDocumentEnvironment { mdalign } { O{} +b }
  {
    \md_model_write:nn {#1} {#2}
    \begin{align}
      #2
      \tl_if_empty:NF \l_md_label_tl
        {
          \label { \tl_use:N \l_md_label_tl }
        }
```

```
    \end{align}
  }
  { }
```

(*End of definition for* `mdalign`*. This function is documented on page* **??**.)

4.7 Term definition command

`\mddefn` `#1` is an equation label (existence is checked), `#2` is the term content (typeset in maths mode), `#3` is the definition text (sidecar only).

```
\NewDocumentCommand \mddefn { m m m }
  {
    \cs_if_exist:cF { r@#1 }
      {
        \msg_warning:nnn { measuredtex } { undefined-label } {#1}
      }
    \iow_now:Nn  \g_md_iow { ----- ~ TERM ~ ----- }
    \iow_now:Nx  \g_md_iow { eqref: ~ \tl_to_str:n {#1} }
    \iow_now:Nx  \g_md_iow { term: ~ \tl_to_str:n {#2} }
    \iow_now:Nx  \g_md_iow { definition: ~ \tl_to_str:n {#3} }
    \iow_now:Nn  \g_md_iow { }
    $#2$
  }
```

(*End of definition for* `\mddefn`*. This function is documented on page* **??**.)

4.8 Semantic term macros

These use conventional $\LaTeX\,2_\varepsilon$ syntax, so we leave `expl3` catcodes temporarily.

```
\ExplSyntaxOff
```

`\nspace`

```
\NewDocumentCommand{\nspace}{m}{%
  \mathrm{#1}%
}
```

(*End of definition for* `\nspace`*. This function is documented on page* **??**.)

`\assoc`

```
\NewDocumentCommand{\assoc}{m}{%
  \mathrm{#1}%
}
```

(*End of definition for* `\assoc`*. This function is documented on page* **??**.)

`\tv`

```
\NewDocumentCommand{\tv}{m}{%
  \mathit{#1}%
}
```

(*End of definition for* `\tv`*. This function is documented on page* **??**.)

\qual

```
118 \NewDocumentCommand{\qual}{m d()}{%
119   \mathrm{#1}\IfValueT{#2}{(\mathrm{#2})}%
120 }
```

(*End of definition for* `\qual`*. This function is documented on page* **??**.)

\mdterm

```
121 \NewDocumentCommand{\mdterm}{o m o d()}{%
122   \IfValueT{#1}%
123     {{}^{\nspace{#1}}}%
124   {#2}%
125   \IfValueT{#3}%
126     {_{\qual{#3}}}%
127   \IfValueT{#4}%
128     {(\assoc{#4})}%
129 }
```

(*End of definition for* `\mdterm`*. This function is documented on page* **??**.)

```
130 \ExplSyntaxOn
131 \endinput
132 ⟨/package⟩
```

Change History

v0.1
General: Initial public version. 1

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The italic numbers denote the pages where the corresponding entry is described, numbers underlined point to the definition, all others indicate the places where it is used.